Randomized trial of pharmacist interventions to improve depression care and outcomes in primary care

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Purpose. The impact of pharmacist interventions on the care and outcomes of patients with depression in a primary care setting was evaluated.

Methods. Patients diagnosed with a new episode of depression and started on antidepressant medications were randomized to enhanced care (EC) or usual care (UC) for one year. EC consisted of a pharmacist collaborating with primary care providers to facilitate patient education, the initiation and adjustment of antidepressant dosages, the monitoring of patient adherence to the regimen, the management of adverse reactions, and the prevention of relapse. The patients in the UC group served as controls. Outcomes were measured by the Hopkins Symptom Checklist, *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition,* criteria for major depression, health-related quality of life, medication adherence, patient satisfaction, and use of depression-related health care services. An intent-to-treat analysis was used.

Results. Seventy-four patients were randomized to EC or UC. At baseline, the EC group included more patients diagnosed with major depression than did the UC group ($p = 0.04$). All analyses were adjusted for this difference. In both groups, mean scores significantly improved from baseline for symptoms of depression and quality of life at three months and were maintained for one year. There were no statistically significant differences between treatment groups in depression symptoms, quality of life, medication adherence, provider visits, or patient satisfaction.

Conclusion. Frequent telephone contacts and interventions by pharmacists and UC in a primary care setting resulted in similar rates of adherence to antidepressant regimens and improvements in the outcomes of depression at one year.

Index terms: Antidepressants; Compliance; Depression; Dosage; Interventions; Patient information; Patients; Pharmacists; Primary care; Quality of life; Toxicity

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More than half of all patients with depression receive their care exclusively from primary care providers (PCPs).[1] The pharmacologic management of depression in primary care has been reported as inadequate with suboptimal clinical outcomes.[2] Depression is frequently treated for an inadequate length of time or with insufficient antidepressant dosages.[3] Patients often discontinue medication because of adverse effects, lack of benefit, or cost. Interventions to improve the quality of care and use of medication have been favorable with the utilization of various nonphysician providers in collaborative care delivery models.[4-6]

The pharmacist's role in the management of depression is evolving. In other chronic diseases, such as hypertension, hyperlipidemia, diabetes, and asthma, pharmacists have had a positive impact on the quality of pa-

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tient care and outcomes of the disease.[7-12] In mental health, pharmacists have the potential to improve the quality of care and outcomes by enhancing compliance, adjustment of medications, and monitoring and managing adverse effects. Pharmacists can also facilitate the use of patient assistance programs if cost is a barrier to treatment.

The purpose of this study was to evaluate the impact of an additional pharmacist's interventions in collaborative care to improve the care of and outcomes for patients with depression at a large university family practice clinic. The goal of this study was to test the effectiveness of pharmacist-delivered multifaceted interventions in collaboration with PCPs and a staff psychiatrist (enhanced care [EC]) versus usual care (UC) in improving quality of care and outcomes to patients diagnosed with a new episode of depression.

Methods

Setting. The University of Washington Family Medical Center (UWFMC) is a primary care clinic in an academic environment. There are 20 faculty family physicians, 18 resident family physicians, 1 behavioral scientist, 1 mental health intern, 1 psychiatrist, 1 psychiatry resident, 1 social worker, 4 physician assistants, 3 registered nurses, 1 clinical pharmacist, 1 pharmacy resident, and 6 medical assistants. The physicians and mental health providers work part-time in the clinic. UWFMC averages 26,000 patient visits annually, and the patient population is 72% Caucasian, 8% Asian, 7% African-American, and 13% other.

UWFMC cares for a complex population of patients representing a broad spectrum of health care needs. Collaboration among the various practitioners maximizes the care provided to patients. The clinical pharmacist has been practicing in this clinic for over 20 years. Responsibilities of the pharmacist span a wide variety of activities, including prescriptive authority under a protocol that allows for the initiation, adjustment, management, and monitoring of pharmacotherapy; triage and care of acute patient problems over the telephone; and smoking cessation, blood pressure monitoring, and disease management. The pharmacist is available for patient education and consultation throughout the day. Teaching responsibilities include lectures to the faculty and resident family physicians throughout the year.

Subjects. Patients were referred to the study by their health care provider if they were diagnosed with a new episode of depression and started on an antidepressant medication. The clinical pharmacist or pharmacy resident screened all referred patients for study eligibility. The initial screening included an assessment for depression using the Primary Care Evaluation of Mental Disorders (PRIME-MD[13]) and two questionnaires to evaluate inclusion and exclusion criteria and alcohol use (Alcohol Use Disorders Identification Test [AUDIT]).[14] Exclusion criteria included (1) age of <18 years, (2) terminal illness, (3) psychosis, (4) recent (within the past 3 months) alcohol (AUDIT score of >8) or substance abuse, (5) two or more suicide attempts, (6) pregnancy or nursing, (7) limited command of the English language, and (8) unwillingness to use UWFMC as a source of care for the next 12 months.

Intervention. The collaborative care intervention employed in the primary care clinic has been described previously.[15] In short, all patients received UC and were encouraged to use available resources (PCPs, pharmacists, nurses, and mental health providers). The EC group received additional follow-up by the clinical pharmacist or pharmacy resident, in conjunction with the PCP and study psychiatrist. The follow-up consisted of weekly telephone calls for the first four weeks, followed by phone contact every two weeks through week 12. During months 4–12, the subjects received a telephone call every other month. Subjects were encouraged to visit their PCP during weeks 4 and 12 (Appendix A).

At each contact, depressive symptoms and medication-related concerns were addressed by the pharmacist. The initial contacts focused on support and education, as well as medication dosage adjustment and the management of adverse effects. Medication refill authorizations were provided, and access to patient assistance programs was facilitated. Other interventions included change in time of dose administrations, change or discontinuation of antidepressants, and provision of additional pharmacotherapy for insomnia or sexual dysfunction, as needed. Appointments with mental health providers were also facilitated (Appendix B).

Outcome measures. Depression symptoms were assessed using the 20-item Hopkins Symptom Checklist (SCL-20) at baseline and 3, 6, 9, and 12 months.[16] The Major Depression module from the Structured Clinical Interview for *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV)* (SCID) was used in combination with the SCL-20 to determine depression remission at 6, 9, and 12 months.[17] The Medical Outcomes Study Short Form 12 (SF-12) was used to assess disability, general health status, pain, and general health perception at baseline and 3, 6, 9, and 12 months.[18]

Antidepressant medication adherence, patient satisfaction, and clinic visits were also evaluated. Two questions, which were used in a previous study, were used to assess patient satisfaction[5]: (1) In the past three months, how would you rate the care that you have received for depression, and (2) How would you rate the care that you have received overall for any treatment in the past three months. A 5-point ordinal scale was used to rate treatment satisfaction, with 1 as "poor" and 5 as "excellent." In addition, medication adherence was measured by asking patients the number of days they took the antidepressant medication in the past month. Previous studies have found excellent agreement between questions regarding the use of antidepressants in the past month and refills from automated pharmacy data (kappa = 0.83–0.90).[5]

Information regarding demographics, socioeconomic status, prior drug treatment for depression, prior psychiatric counseling or psychotherapy, panic disorder, dysthymia, and neuroticism was also assessed at baseline.

Statistical analysis. Power analysis and sample-size calculations have been described previously.[15] A generalized linear model was used to assess outcomes in the intent-to-treat analysis. Generalized estimating equations were used to account for repeated measures on each subject. For dichotomous outcomes, the binomial family of distributions with a logit link was used; for continuous outcomes, the normal family of distributions with an identity link was used. If the outcome was measured at baseline, the model parameters for the effect of treatment group were restricted to be equal at baseline. Baseline SCID score was included in all models as a covariate since there was a significant difference between groups at baseline. The effects of treatment group, time, and treatment by time interaction were included in all models. The methodology of generalized estimating equations can accommodate missing data, assuming that they are randomly missing.[19] The available data were used for parameter estimation and hypothesis testing.

For health care utilization outcomes, the Wilcoxon rank sum test was used to assess group differences.

Results

A total of 106 patients were considered for inclusion in the study by resident physicians, attending physicians, clinical pharmacists, and other providers, such as physician assistants and counselors. Of these patients, 17 were ineligible and 8 refused to participate. In addition, 5 eligible and consenting participants were lost because a baseline interview was not conducted within seven days from the date antidepressant medication was prescribed; 2 patients refused the baseline interview. Seventy-four patients (70%) were successfully enrolled in the study; 41 were randomized to EC and 33 to UC.

Of the 74 patients enrolled in the study, 71 (96%) completed the 3-month follow-up, 70 (95%) completed the 6- and 9-month follow-ups, and 69 (93%) completed the 12-month follow-up. Of the 41 EC patients, 37 (90%) versus 30 (91%) UC patients completed all four follow-ups ($\chi^2_1 = 0.1$, $p = 0.99$).

Patient characteristics. The majority of subjects were women (57%) and Caucasian (78%), and the mean ± S.D. age was 39 ± 13.5 years. Over half were employed at baseline, and 93% were high-school graduates (Table 1). The majority of subjects had a yearly household income over $30,000, and 95% had some form of health insurance at baseline. Approximately 43% had previously received antidepressant medications for depression, and 46% had received counseling or psychotherapy in the past for depression. There were no significant differences on any demographic or clinical variables between the two groups (Table 1). At baseline, more of the patients in the EC group had been diagnosed with major depression (*DSM-IV*) than in the UC group (Table 2).

EC interventions. Of the 41 patients randomized to EC, 14 (34%)

Table 1.
Baseline Demographics of Enhanced Care versus Usual Care Patients

Characteristic	No. (%) Patients Total ($n = 74$)	Enhanced Care ($n = 41$)	Usual Care ($n = 33$)	p
Women	57 (77)	34 (83)	23 (70)	0.18
Mean ± S.D. age (yr)	38.7 ± 13.5	38.2 ± 13.8	39.4 ± 13.4	0.71
Nonwhite	16 (22)	9 (22)	7 (21)	0.94
Currently employed	53 (72)	30 (73)	23 (70)	0.74
Completed high school	69 (93)	38 (93)	31 (94)	1.00
Annual household income <$30,000	19 (26)	12 (29)	7 (21)	0.36
Health insurance status				0.41
Private or managed care	58 (78)	33 (80)	25 (76)	
Medicare or Medicaid	8 (11)	5 (12)	3 (9)	
Other	3 (4)	2 (5)	1 (3)	
None	5 (7)	1 (2)	4 (12)	
Smoking status				0.08
Current	12 (16)	5 (12)	7 (21)	
Past	20 (27)	8 (20)	12 (36)	
Never	42 (57)	28 (68)	14 (42)	
Clinical status				
Panic disorder	14 (19)	9 (22)	5 (15)	0.43
Mean ± S.D. NEO neuroticism score[a]	11.7 ± 5.8	12.4 ± 6.1	11.0 ± 5.5	0.31
Dysthymic disorder	39 (53)	23 (56)	16 (48)	0.40
Prior antidepressant for depression	32 (43)	20 (49)	12 (36)	0.28
Prior counseling or psychotherapy	34 (46)	17 (41)	17 (52)	0.39

[a]NEO (neuroticism, extraversion, openness) Personality Inventory.

Table 2.
Baseline Clinical and Economic Outcomes of Enhanced Care versus Usual Care Patients[a]

Characteristic	Total ($n = 74$)	Enhanced Care ($n = 41$)	Usual Care ($n = 33$)	p
Mean ± S.D. SCL-20 score	1.79 ± 0.07	1.83 ± 0.10	1.75 ± 0.10	0.55
No. (%) with SCID major depression	30 (42)	21 (53)	9 (28)	0.04
Mean ± S.D. SF-12 Index (physical) score	50.9 ± 1.2	49.6 ± 1.6	52.6 ± 1.6	0.68
Mean ± S.D. SF-12 Index (mental) score	28.4 ±1.2	28.0 ± 1.6	29.0 ± 1.7	0.20
No. (%) who had counseling or psychotherapy in the past 3 mo	40 (54)	24 (59)	16 (48)	0.39
No. (%) who had been hospitalized in the past 3 mo	2 (3)	1 (2)	1 (3)	1.00
No. (%) who had an ER visit in the past 3 mo	10 (14)	7 (17)	3 (9)	0.32
No.(%) who had missed work or school in the past 3 mo	40 (54)	22 (54)	18 (55)	0.94

[a]SCL-20 = 20-item Hopkins Symptom Checklist, SCID = Structured Clinical Interview for *Diagnostic and Statistical Manual, Fourth Edition,* SF-12 = Medical Outcomes Study Short Form 12, ER = emergency room.

completed all 13 sessions, 33 (80%) completed at least 10 sessions, and 39 (95%) completed at least 6 sessions. The median number of clinical pharmacist interventions per patient was 11 (range, 3–13). The median time required for the pharmacist to complete an intervention was 15 minutes (range, 5–50 minutes). Approximately 109 hours of pharmacist time were spent conducting 440 interventions during the one-year study period. Of the 440 interventions performed, 365 (83%) were conducted over the telephone. The 75 inclinic interventions occurred at various times during the study, either by the pharmacist or PCP. Only 14% of all inclinic interventions occurred at the recommended weeks 4 and 12. The majority of the inclinic interventions (35%) occurred at week 1. The range of inclinic interventions per subject was zero to five. Six subjects (8%) never came to the clinic for an intervention at any time during the study period. All EC patients were included in the analyses, regardless of whether they completed the interventions.

Medication adherence. Based on the self-reported telephone interview data, there was no difference between treatment groups in overall antidepressant medication adherence (χ^2_1 = 0.01, p = 0.91). The association between treatment groups and medication adherence was similar in subgroups defined by follow-up. The EC and UC groups demonstrated similar trends in reporting adherence to antidepressants (defined as use of antidepressants for at least 25 of the past 30 days) at the 3-month (85% versus 81%), 6-month (78% versus 73%), 9-month (48% versus 67%), and 12-month follow-ups (59% versus 57%) (Figure 1).

Patient satisfaction. There was no overall difference in satisfaction with depression care (χ^2_1 = 1.75, p = 0.19) or overall health care (χ^2_1 = 0.51, p = 0.48) between groups. Both the EC and UC groups reported good or excellent quality of care at the 3-month (78% versus 58%), 6-month (88% versus 73%), 9-month (78% versus 77%), and 12-month (80% versus 77%) follow-ups. More subjects in the EC group than in the UC group reported good or excellent quality of depression care during the first 6 months of the study, when the interventions were more frequent.

Changes in outcomes. Mean SCL-20 and SF-12 mental health scores and the number of patients with a *DSM-IV* diagnosis of major depression improved from baseline in both groups during the 12-month follow-up period (Figures 2–4). However, the overall difference between the groups during that follow-up period was not significant for the main study outcomes: mean SCL-20 score (χ^2_1 = 0.01, p = 0.92), mean SF-12 mental health score (χ^2_1 = 0.54, p = 0.46), diagnosis of major depression (χ^2_1 = 0.98, p = 0.32), and mean SF-12 physical health score (χ^2_1 = 1.76, p = 0.18). Subgroup analyses of the patients with major depression found no significant difference in SCL-20 scores between treatment groups (χ^2_1 = 0.01, p = 0.94). Both the EC and UC groups clinically improved in depression symptoms (defined as a 50% or more decrease in SCL-20 score from baseline) at the 3-month (52% versus 64%), 6-month (72% versus 67%), 9-month (75% versus 73%), and 12-month follow-ups (72% versus 80%). However, the number with a 50% or more decrease in SCL-20 score during the study period did not differ between groups (χ^2_1 = 0.75, p = 0.39).

Health care visits. Table 3 summarizes the self-reported visits to health care providers during follow-up. Using the Kruskal-Wallis test, subgroup analyses of specific health care providers found no difference between treatment groups in the number of visits to all health care providers (χ^2_1 = 0.0003, p = 0.99), physicians (χ^2_1 = 0.02, p = 0.88), psychiatrists or psychologists (χ^2_1 = 0.0003, p = 0.99), emergency rooms (χ^2_1 = 1.21, p = 0.27), counselors or other mental health providers (χ^2_1 = 1.07, p = 0.30), and alternative medicine providers (χ^2_1 = 0.57, p = 0.45).

Figure 1. Percentage of patients adherent to antidepressant regimen (25 days or more of use in past 30 days) over 12 months of follow-up. Data for the enhanced care group and the usual care group are indicated by a solid line and a dashed line, respectively.

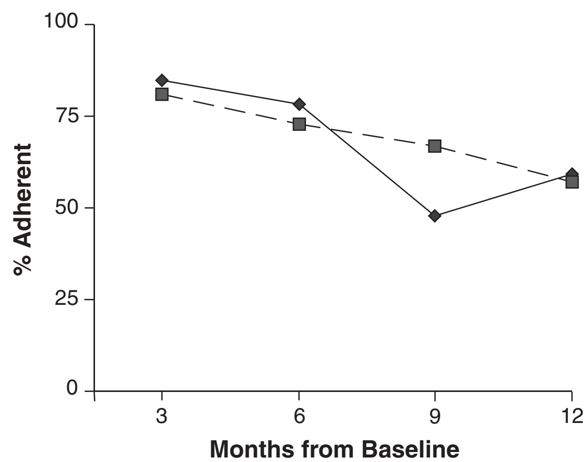
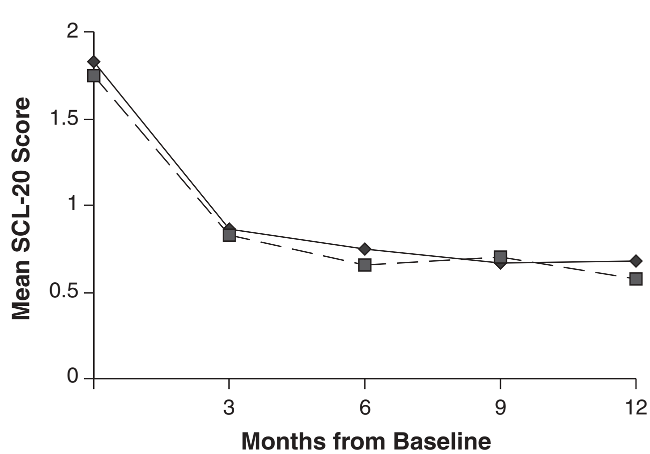


Figure 2. Mean score on 20-item Hopkins Symptom Checklist (SCL-20) over 12 months of follow-up. Data for the enhanced care group and the usual care group are indicated by a solid line and a dashed line, respectively.



Discussion

This study found substantial improvement in depression symptoms from baseline at 3, 6, 9, and 12 months in both the EC and UC groups. The majority of patients reported a high degree of satisfaction with care, and adherence rates to antidepressant medication were surprisingly high throughout the study period. These results are encouraging given that the management of depression in primary care is often reported as inadequate, with suboptimal clinical outcomes and poor adherence rates.[2]

Unlike other published studies on collaborative care models, we did not find any difference in outcomes between the EC and UC groups. Previous research in primary care has found the integration of pharmacists and other health care providers to be favorable in the treatment of depression. In a randomized controlled trial conducted in a large primary care clinic of a health maintenance organization, Katon et al.[20] found that increasing the intensity and frequency of visits to the PCP and study psychiatrist improved depression outcomes, adherence, and patient satisfaction among EC patients. Study outcomes were assessed at one, four, and seven months from randomization. The difference in outcomes between EC and UC groups was predominantly found among those patients classified as having major depression, according to *DSM-IV*,[17] at baseline. In patients with minor depression, the potential difference in clinical outcomes between EC and UC patients was limited by a favorable response in the UC group. Stratifying our data by depression severity (major versus minor depression) did not change the findings.

A second randomized trial by Katon and colleagues[5] studied the effects of a multifaceted intervention that included frequent visits (four to six) to a psychologist, provision of educational material, telephone contacts, and brief psychotherapy on major depression outcomes. Similar to their prior study,[20] they found a significant improvement in medication adherence to antidepressant medication and depression severity over time compared with the UC group. A large randomized trial conducted within a managed care setting found that a multifaceted quality improvement program, which included training local experts and nurses to provide clinician and patient education, medication follow-up by trained nurses, and access to mental health providers, was associated with improved quality of life, depression symptoms, and retention of employment.[4] The investigators did not observe an increase in medical visits over the 12-month follow-up period.

In a prospective, nonrandomized study of a collaborative pharmacy practice model to manage patients with mild to moderate depression, Finley et al.[21] reported improved medication adherence over a 6-month period and a decreased num-

Figure 3. Mean mental health score on the Medical Outcomes Study Short Form 12 (SF-12) over 12 months of follow-up. Data for the enhanced care group and the usual care group are indicated by a solid line and a dashed line, respectively.

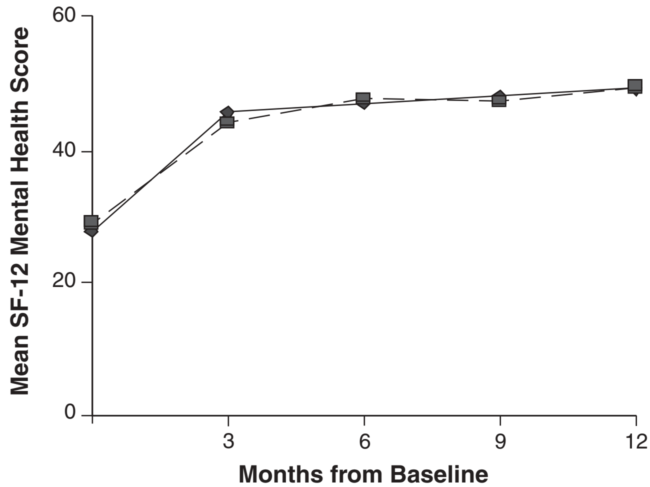


Figure 4. Percentage of patients meeting *Diagnostic and Statistical Manual, Fourth Edition (DSM-IV)* criteria for major depression over 12 months of follow-up. Data for the enhanced care group and the usual care group are indicated by a solid line and a dashed line, respectively.

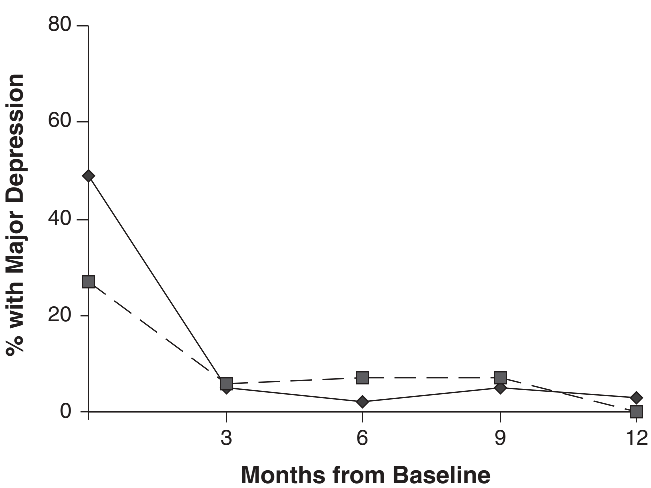


Table 3.
Summary of Clinic Visits over One Year of Follow-up[a]

Outcome	Median (Range) Enhanced Care ($n = 41$)	Median (Range) Usual Care ($n = 33$)	p
All provider visits	9 (0–42)	9 (0–60)	0.99
PCP visits	4 (0–21)	5 (0–17)	0.88
Psychiatrist visits	0 (0–6)	0 (0–6)	0.99
Emergency room visits	0 (0–2)	0 (0–2)	0.27
Counselor visits	0 (0–33)	1 (0–32)	0.30
Alternative medicine provider visits	0 (0–29)	0 (0–43)	0.45

[a]PCP = primary care provider.

ber of subsequent visits to PCPs and psychiatrists over a 12-month period. The pharmacist intervention was similar to that used in our randomized trial, and, although encouraging, the nonrandomized design of Finley et al.'s study may have introduced selection bias.

Possible explanations for the similar outcomes of the EC and UC groups in our study may be high-quality care from PCPs, the established role of the clinical pharmacist, and the ease of access to the pharmacist and various mental health providers in the clinic. The presence of mental health services and provider education may have increased the knowledge and skill of all providers in caring for patients with depression over the years. UWFMC has a wealth of resources, compared with many primary care practices, to help with the management of depression. Both groups were encouraged to use all available resources. With four mental health providers, a pharmacist, and a pharmacy resident, patients' access to these resources cannot be ignored. Patients may have interacted or utilized the available resources without the provider's knowledge of their treatment group. A patient in the UC group may have asked for a refill or posed questions or discussed adverse effects with the pharmacist. This care is considered standard and may not have been documented as an intervention in the UC group. Telephone or e-mail communication between physicians and patients may have occurred without the proper documentation of this intervention. The high standard of care in this clinic may have affected the results. A Hawthorne effect cannot be ruled out, since patients in the EC and UC groups received follow-up telephone calls from a research assistant at 3, 6, 9, and 12 months.

Most primary care studies of major depression have found that only about 40% of patients in the UC group have a 50% or greater reduction in SCL-20 scores.[5,20,22] Our results were much higher, which limits our ability to demonstrate intervention differences. This contributes to the possibility that this clinic provides enhanced UC because it has a clinical pharmacist on staff, as well as psychiatrists, psychologists, and a social worker.

We also cannot rule out the possibility of a spillover effect into UC, but previous experience has shown a surprisingly small effect, causing only small decrements in the otherwise robust differences between the intervention and control groups.[23] Failure to comply with the study protocol may have also contributed to a lack of difference in outcomes. Although the EC treatment protocol was adequately followed with respect to telephone interventions, most patients (86%) did not adhere to the scheduled clinic visits at weeks 4 and 12. This could be due to the intense follow-up by telephone or not reinforcing the necessity to return to the clinic. The majority of clinic visit interventions (35%) occurred during week 1.

The adherence to antidepressant medication in both groups was remarkable throughout the 12 months of the study. The Agency for Health Care Policy and Research (AHCPR) recommends that patients being treated for a major depressive episode complete 4–9 months of antidepressant therapy after symptoms resolve to prevent relapse.[24,25] Adherence rates for 6 months of antidepressant therapy are quite variable and range from 4% to 40% in natural settings.[21] In this study, the 6-month adherence rate exceeded 70% in both groups. Patient adherence to antidepressants was measured by self-report in this study and was not validated by pharmacy or medical records.

There are several limitations that should be considered when interpreting our study results. This was a randomized controlled trial, and although every effort was made to work within the normal constraints of a busy primary care setting, generalization to other settings should be done with caution. The relatively homogeneous study sample (i.e., predominantly white, high-school-educated, young, insured women) may limit the generalizability of our findings to other populations. Data collection was conducted via telephone interviews and thus subject to recall bias.

Another limitation of this study is the possibility that self-reported adherence of antidepressant use was inaccurate. While there are no published data on the validity of self-reported adherence to antidepressants, a few validation studies of self-reported antidepressant use have been conducted. Cotterchio and colleagues[26] compared the accuracy of self-reported antidepressant use with physicians' records for adult female cancer patients and controls. The study reported 80% agreement (kappa = 0.60; 95% confidence interval [CI], 0.47–0.74) for overall lifetime antidepressant use. A more recent validation study, conducted among participants of a randomized trial of depression detection and treatment, found high concordance between self-reported use of antidepressants and claims data (agreement = 85%, kappa = 0.69).[27] Relative to the claims data, the self-report measure had a sensitivity of 74% and a specificity of 94%. One of our investigators, however, did not find self-reported antidepressant use data, which was collected among older women participating in a breast cancer case–control study, to be very accurate when compared with automated pharmacy records.[28] Among those classified by the pharmacy records as using antidepressants during the six months prior to the reference date, 64% of cases (95% CI, 57–70%) and 66% of controls (95% CI, 59–72%) recalled having used antidepressants during this period. Specificity estimates approached 100%, indicating little overascertainment of antidepressant use. While these studies provide some indication of the adequacy of self-reported adherence, we recognize that adherence and overall use are different measures. In addition, the validation studies of antidepressant medication use have been conducted among adults over 65 years of age. It has been suggested that adults over age 65 do not recall medication names as well as younger adults; therefore, the results of these validation studies may not be generalizable to our study population.

Previous research demonstrates that pharmacists working as members of interdisciplinary primary care teams can positively affect patient outcomes, satisfaction, quality of medication prescribing, and the delivery of care.[29-31] Although our study did not show a difference in the improved outcomes between EC and UC groups, it is hoped that this project may encourage other institutions to study collaborative care practice models to improve depression treatment. The feasibility and success of such models are likely to vary with the availability of clinic resources, patient populations, and clinic settings.

Conclusion

Frequent telephone contacts and interventions by pharmacists and UC in a primary care setting resulted in similar rates of adherence to antidepressant regimens and improvements in the outcomes of depression at one year.

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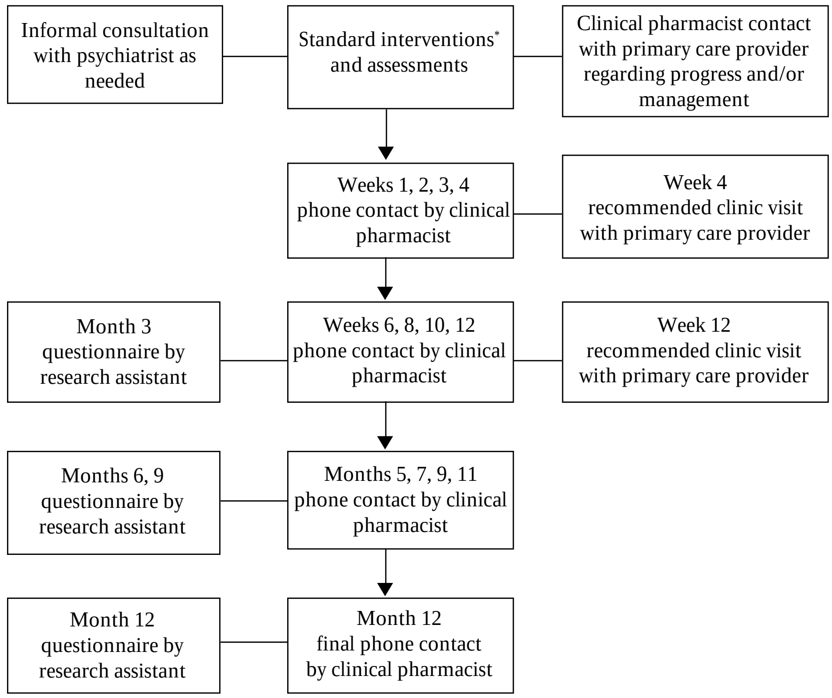
Appendix A—Flow chart of standard intervention and follow-up[a]



*See intervention list. Scheduled interventions and follow-ups are approximate and vary according to patient/provider schedules.
[a]Reprinted, with permission, from reference 15.

Appendix B—Standard interventions performed by clinical pharmacist[a]

Main Interventions	Specifics and Motivation of Intervention
Reassurance	Support during initiation of therapy
	Education about medication or depression
	Encourage compliance
	Reinforce goals and motivate
Dose adjustment	Initial titration phase
	Adverse effects
	Efficacy
	Discontinuation taper
Dose time change	Morning to evening if sedated, drowsy
	Evening to morning if sleep disturbances
	Split dose
	Compliance
Change of antidepressant	Not effective
	Adverse effect
Discontinuation of antidepressant therapy	Adequate trial
Other	Add agent for insomnia
	Add agent to treat sexual dysfunction
	Weekend drug holidays to treat sexual dysfunction
	Refill medication
	Facilitate physician/psychiatrist appointments

[a]Reprinted, with permission, from reference 15.

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